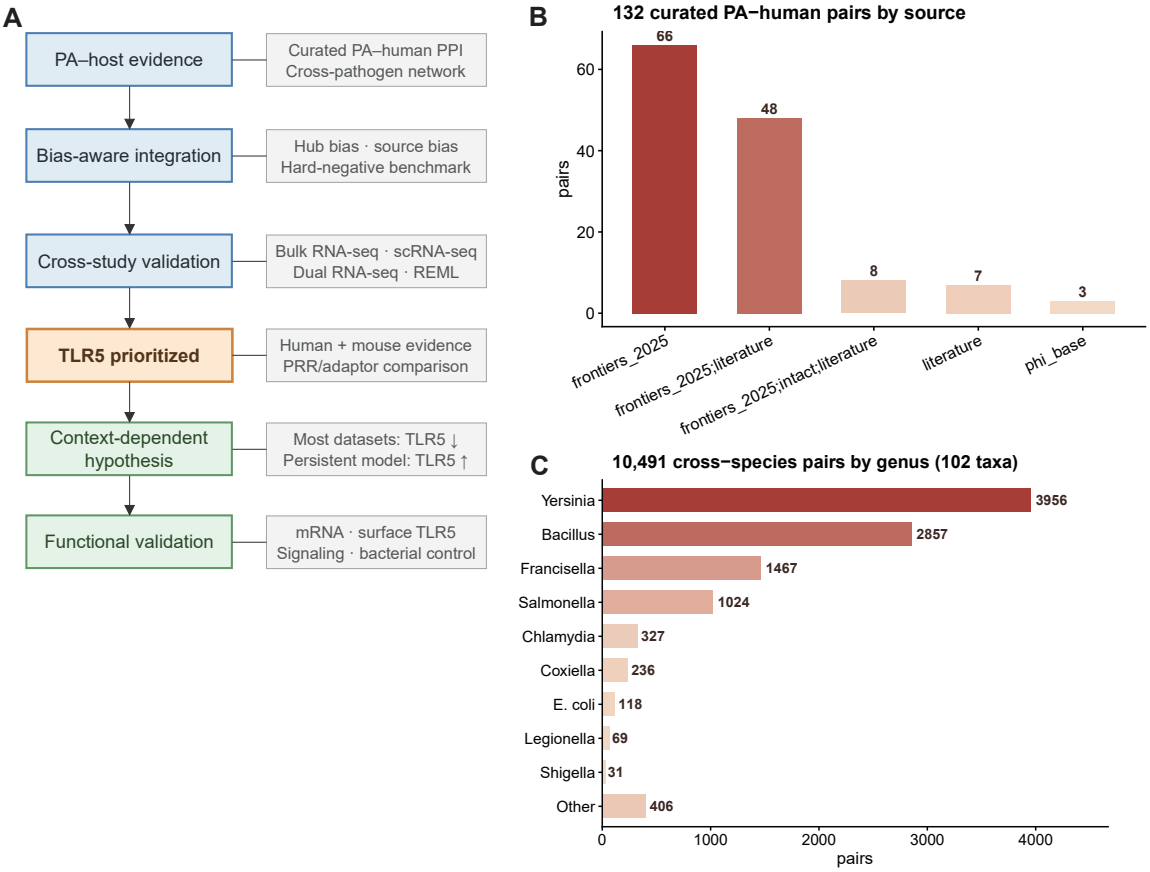

A
PA–host evidence
Curated PA–human PPI
Cross-pathogen network
Bias-aware integration
Hub bias · source bias
Hard-negative benchmark
Cross-study validation
Bulk RNA-seq · scRNA-seq
Dual RNA-seq · REML
TLR5 prioritized
Human + mouse evidence
PRR/adaptor comparison
Context-dependent hypothesis
Most datasets: TLR5 ↓
Persistent model: TLR5 ↑
Functional validation
mRNA · surface TLR5
Signaling · bacterial control
B
132 curated PA−human pairs by source
pairs
0
20
40
60
66
48
8
7
3
frontiers_2025
frontiers_2025;literature
frontiers_2025;intact;literature
literature
phi_base
C
10,491 cross−species pairs by genus (102 taxa)
Yersinia
Bacillus
Francisella
Salmonella
Chlamydia
Coxiella
E. coli
Legionella
Shigella
Other
3956
2857
1467
1024
327
236
118
69
31
406
0
1000
2000
3000
4000
pairs